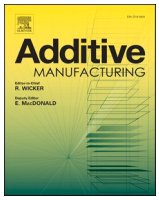
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Design, testing, and application of an open-source powder material extrusion 3D printer

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ABSTRACT

Powder material extrusion (PME) additive manufacturing (AM) is a convenient and practical method to study novel materials by circumventing the need for filamentation or compounding of materials. Avoiding additional processing steps can be an enabler for research with exploratory materials or those that otherwise display thermal instabilities. In this work, we present the design, development, and testing of an open-source PME printer. The open-source PME 3D printer achieves print quality on par with commercial material extrusion 3D printers in terms of dimensional accuracy, print quality, and mechanical performance. Additionally, we demonstrate this system to be versatile and robust through printing of recycled materials, polymer composites, polymer blends, and functional polymers with thermally sensitive moieties. The broad range of build materials illustrates the diverse capabilities accessible with this system, which enabled access to properties and functions such as phosphorescence, ferromagnetism, shape memory, and mechanochromism. By improving the accessibility of PME AM and demonstrating its versatility, we hope to enable others to explore novel material systems.

1. Introduction

Additive manufacturing (AM), also commonly referred to as 3D printing (3DP), encompasses technologies for producing complex objects from computer aided design models [1]. Object function is instilled through material and geometry choices that impart unparalleled design freedom. This freedom has made AM technologies popular in academia and industry [2,3] resulting in new structure property relationships [4,5], novel 4D functionalities [6–10], smart materials capabilities [11–14], biomimetic materials [7,15–18], and revelations in personalized healthcare [19–21]. Such innovations are made possible through advancements in hardware, material scope, and software – collectively the pillars of the AM ecosystem.

Material extrusion (MEX) AM is a subclass of AM characterized by the controlled dispensing of softened material through a nozzle [1]. The most common type of MEX is fused filament fabrication (FFF) which utilizes a rastering hot end extruder to dictate the deposition of melted thermoplastic polymer around a build platform. FFF systems have undergone extensive advancements over the past decade yielding inexpensive, easy to operate, modifiable, multimaterial-capable, and high-performance printers [22–24]. With these advancements,FFF systems have become widely used for fabrication of end-user products, in prototyping, and in research settings [3].

The materials compatible for FFF are predominately thermoplastics that must be formed into spooled filaments [24,25]. Common materials for FFF printing include polylactic acid (PLA), polycarbonate (PC), polyethylene terephthalate glycol (PETG), and acrylonitrile-butadiene-styrene (ABS). Although these materials are suitable for many applications, the desire for higher performing and specialty materials is ever growing [24–26]. One challenge with studying novel material systems in FFF is the non-trivial process for forming filament spools. This process requires an understanding of rheology, material flow, and heat transport, as well as specialized equipment [25]. Additionally, research-scale filament extruders can be costly and often require quantities of material that are difficult to produce in a purely research setting, which makes experimental iteration cumbersome. Furthermore, the additional filamenting process can be a challenge for materials that exhibit thermal sensitivity making direct extrusion without filamenting appealing [27,28].

Direct MEX printing of granulated material as opposed to filament

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would address many of the issues outlined above [29]. This has motivated exploration of MEX methods such as screw-assisted material extrusion printers [30]. These printers offer several possible advantages including broader material scope, larger design freedom, and reduced feedstock cost in comparison with those that require precisely formed filament [24,26,29,31–33]. Screw-assisted printing technologies have shown significant potential, especially in their ability to continuously print a diverse range of granulated materials. These include directly recycled substances [32,34,35], biological materials [33,36], and metals and ceramics [34,37]. Notably, many screw-assisted systems are optimized for processing pellets (granulated materials larger than 1 mm), which offers cost advantages due to the direct use of bulk-distributed materials. However, a limitation arises when printing composites with varied granule sizes. In such cases, an initial compounding step followed by pellet formation is necessary. In contrast, using finer powders (less than 1 mm in size) in a powder material extrusion (PME) printer could directly create composite objects, bypassing the need for compounding. This approach highlights the unique capabilities of PME printers in handling a wider range of material sizes and types.

Although there are various commercial screw-assisted printers available for granular printing, most face significant barriers to widespread adoption, primarily due to their cost and complexity. In contrast, a low-cost, customizable, and versatile open-source printer could offer a more accessible option. Such a printer would allow researchers to experiment with a broader range of materials and functionalities without the constraints of filament-based systems. While several groups have developed printers aimed at this goal, challenges remain [26,29,30,38]. Most notably, existing designs are often proprietary, struggle to match the print quality of FFF methods, and have not fully tapped into the potential of novel build materials.

To this end, we introduce a low-cost, open-source PME 3D printer and demonstrate print characteristics on par with those achieved with commercial FFF printers. Additionally, we showcase facile preparation and printing of recycled plastics, polymer blends, polymer composites, and specialty polymers. We briefly explore the useful properties of these material combinations and discuss future opportunities. By doing this, we showcase the expanded design freedom a completely open-source and modular PME system offers and create a versatile, robust, and cost-effective platform for others to explore novel material systems.

2. Technical design

In the development of a PME 3D printer, we focused on establishing a cost-effective, open-source system with performance comparable to commercial FFF printers. The Ender 3 desktop 3D printer by Creality was selected as the base 3D printer, primarily for its open-source architecture (Fig. 1). This choice facilitated the integration of a custom print head while also allowing the use of the original Ender-3 parts in the design, effectively reducing both costs and complexity.

The primary challenge in 3D printing, and a key focus for the PME printer, was achieving consistent material delivery. In both PME and FFF systems, this requires maintaining a continuous and even flow of material into the printer's hot end. The initial design iteration, taking inspiration from Boyle et al. [26], featured a funnel and a single screwhead extruder powered by a NEMA 17 stepper motor. This configuration, however, resulted in inconsistent extrusion patterns, as evidenced by a coefficient of determination (R^2) of 0.9425 and gradual decrease in powder flow over time (Fig. 2). To address this, the design was revised to include an additional lateral chamber for powder storage and an extra feed screw, significantly improving powder delivery consistency, reflected in an improved R^2 value of 0.9988 (Fig. 2).

Further refinements led to the incorporation of a 1:3 ratio synchronous gear belt drive system (Fig. 3 A). This system was engineered to compact the material transfer system's size and amplify the motor's torque, optimizing space and enhancing power transmission efficiency.

In adapting the Ender-3's hotend for the PME printer, removal of the Bowden tubes created space for a 4 mm diameter pathway, originally designed for the Bowden tube, to accommodate the extrusion drill bit

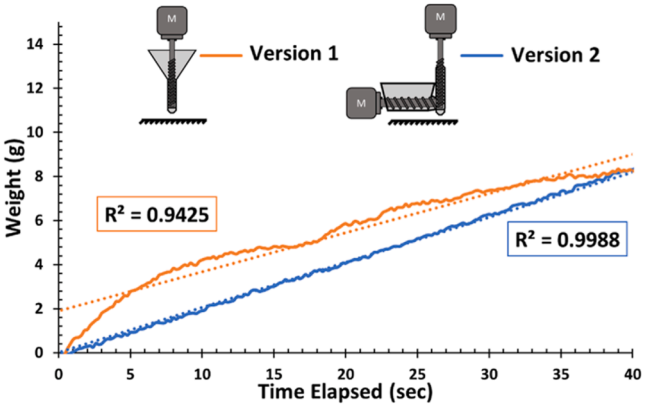


Fig. 2. Powder transfer comparison between the original design (orange) and the version 2 design (blue). Dotted lines and R^2 values show the linear best fit for each curve.

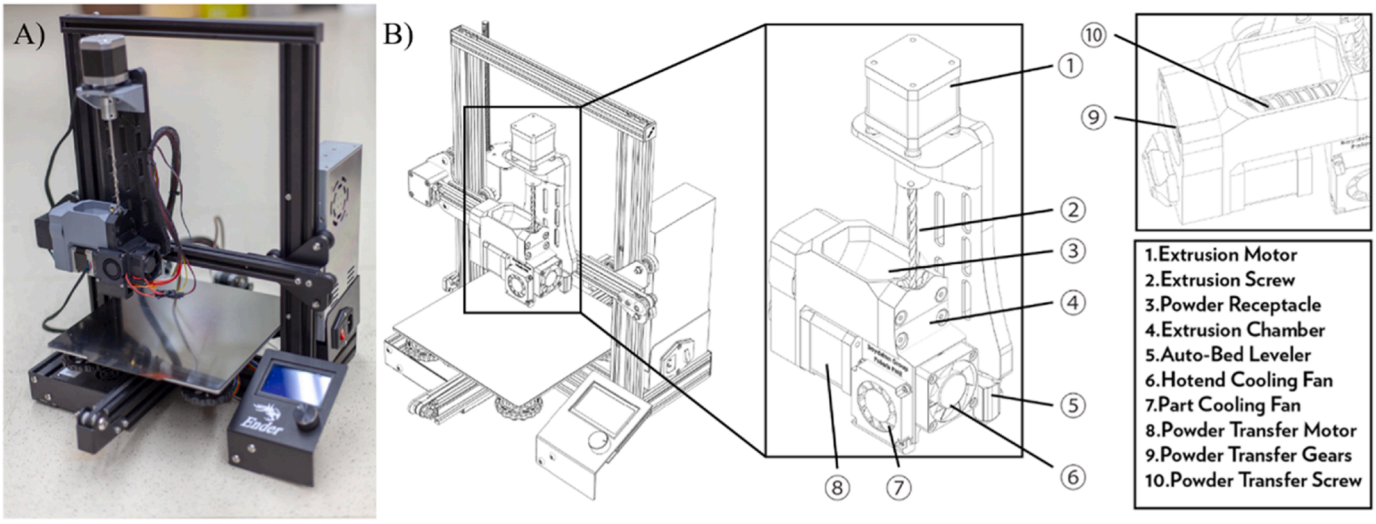


Fig. 1. A) PME printer build B) PME printer design with the print head parts identified.

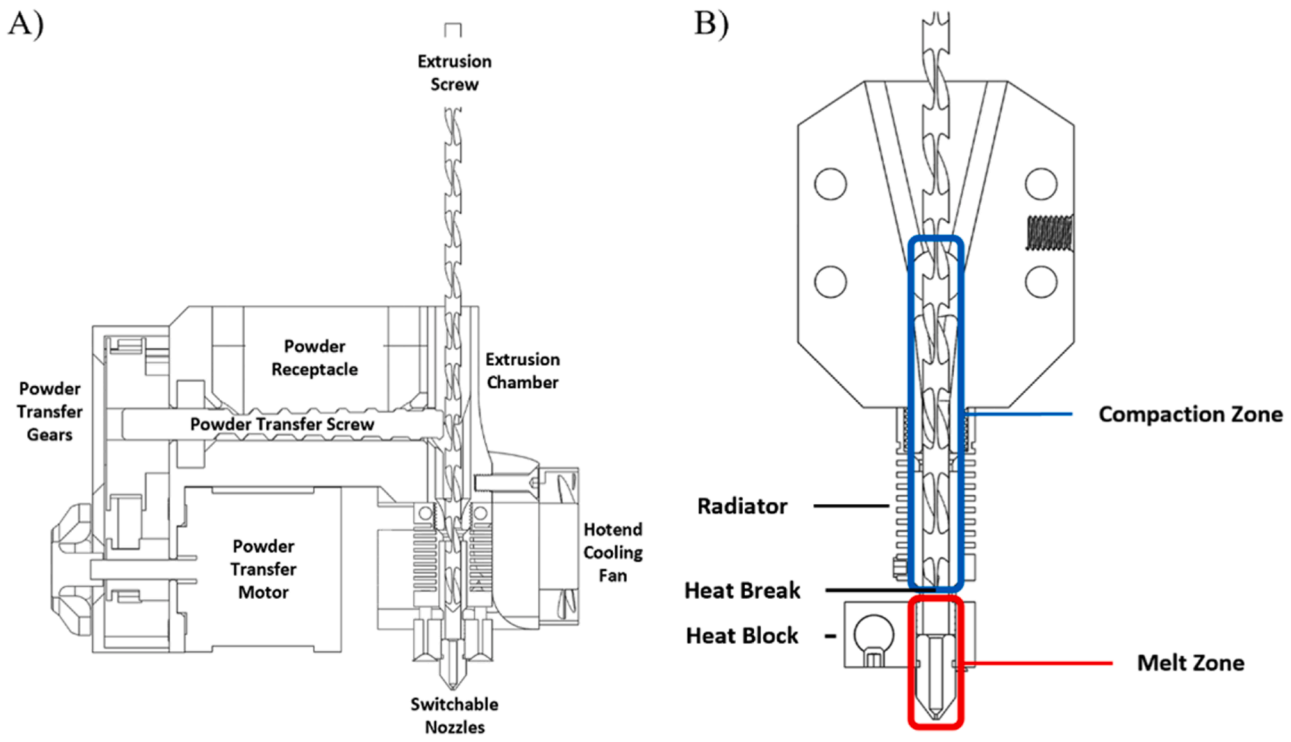


Fig. 3. A) Cross section of open-source PME printer head with main components labeled. B) Cross section of the hot end design showing the compaction zone and melt zones.

(Fig. 3B). Crucially, this modification maintained much of the hotend's original structure, enabling the reuse of the existing heater, thermistor, and, importantly, the nozzles. This reuse of original components not only eased integration into the PME printer but also contributed to cost savings and waste reduction.

The modification of the hotend necessitated the design of a new fan shroud to accommodate both the part cooling fan and the hotend cooling fan (Fig. 3A). Additionally, a mounting mechanism for the NEMA 17 extrusion stepper motor was engineered to drive the extrusion screw. A flexible coupler was employed to connect the motor's shaft to the extrusion screw (Fig. 1B). This setup is compatible with various stepper motors including different sizes and gear ratios to accommodate a wider range of extrusion speed and strength.

The entire assembly, including the modified hotend and motor components, was then mounted onto a 3D printed scaffold. This scaffold provided a stable and precise platform for the components, ensuring proper alignment and functionality. Furthermore, a custom stepper motor controller was developed to manage the operation of the material transfer motor, allowing for precise control over the material extrusion process.

Fig. 1 shows the entire printer and print head design. The design consists of a side powder receptacle which facilitates the consistent transfer of powder from the receptacle to the extrusion chamber by a powder transfer screw driven by the powder transfer motor and powder transfer gears (Fig. 1B). Once the powder is in the extrusion chamber, it is moved toward the hot end by the extrusion screw which also compresses the powder (Fig. 1B). The powder softens and melts as it nears the hot end tip and is extruded onto the build platform (Fig. 3B). The final cost at the time of publication was comparable to entry-level FFF 3D printers (Table S1).

3. Materials and methods

3.1. Materials

SainSmart thermoplastic polyurethane (TPU) filament, Overlook polylactic acid (PLA) filament, MSNJ polycaprolactone (PCL) filaments, strontium aluminate-based glow powder, and iron oxide powder were purchased from Amazon. Spiropyran difunctional initiator was prepared as previously reported[39,40]. Stannous octoate (92.5–100 %), ε-caprolactone (97 %), and tetrahydrofuran (>99.9) were purchased from Sigma Aldrich. HDPE for recycling via PME 3DP was obtained from laboratory scintillation vial caps and soda bottle caps.

3.2. Thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), and dynamic mechanical analysis (DMA)

TGA was conducted on a TA TGA Q50 under nitrogen from room temperature to 600 °C with a heating rate of 10 °C/min. DSC was conducted on a TA DSC Q200 calorimeter under air. Samples were sealed in a Tzero aluminum pan and lid. Testing was run under a heat-cool-heat cycle with 10 °C/min heating and cooling rates. The first cooling and second heating are reported. The three-point bending test DMA experiments were performed on a TA Instruments RSA III in a 3-point bending configuration performed at 0.01 % strain and 1 Hz.

3.3. PME printing

To print non-powdered thermoplastic materials with the PME printer, the given material was ground into powder using a coffee grinder and by periodically adding dry ice to embrittle the polymers. The resulting powder was separated through a 35 mesh sieve and the powder was recovered and stored in a desiccator to remove residual moisture. Although more elaborate operation of the PME printer can be found in the Supporting information PME Build Guide, in brief, stl files were loaded into Cura and sliced using the desired print parameters. The

Gcode file was loaded onto the printer. Powdered material was then fed into the material hopper and the powder transfer was started. Once material was being fed into the extruder chamber, the print was started. All prints were done with a 0.6 mm nozzle unless directly indicated otherwise.

3.4. Particle size determination

Particle size distribution was determined through optical microscopy followed by image analysis. For measurement, a small amount of powder was dispersed in water and sandwiched between two glass slides. The glass slide was imaged using a Nikon Eclipse Ti-2 microscope. The microscopy image was then analyzed using imageJ to determine particle size distribution. The resulting powder had an average particle size of 140 ± 40 μm (Fig. S5).

3.5. Scanning electron microscopy (SEM)

The Prep-Leica ACE600 was used to sputter coat samples with a 10 nm thick layer of platinum with settings of 30 mA, a 2.5×10^{-2} mbar operating pressure, working distance of 50 mm, and sputter rate of 0.13 nm/s. Samples were then imaged on a Zeiss GeminiSEM 450 scanning electron microscope.

3.6. Tensile testing

Tensile testing was performed on an MTS universal testing machine. ASTM D638 type V dogbone samples were prepared for each material. For rigid materials (PLA and PCL), a strain rate of 10 mm/min was used. For soft materials (TPU), a strain rate of 100 mm/min was used. Three replicates were performed for each material and the average and standard deviation of those measurements reported.

3.7. Bending test

ASTM D638 type V dogbone samples were printed with various ratios of TPU to PCL. The parts were then submerged in 60 °C water for 1 min. After 1 min, the samples were bent to an angle of 180° around a 3 mm diameter post and held in that position until completely cooled. The samples were resubmerged in 60 °C water and their relaxation tracked by video. Angles were determined by analyzing the resulting video frames using MATLAB.

3.8. Synthesis of PCL-spiropyran-PCL

The mechanochromic PCL, P1, was prepared as previously reported (M_n = 37.2 kDa and $Đ$ = 1.45) (Fig. S15) [39,41]. Gel permeation chromatography (GPC) was performed using an Agilent Technologies 1260 Infinity II pump with two inline Agilent Plgel (part #: PL1113–6300) columns, Wyatt Technology mini-DAWN light scattering, and Agilent 1260 Infinity II RI detectors, using uninhibited HPLC grade THF as the mobile phase.

4. Validation

Although PME and FFF platforms can often use similar feedstock materials, FFF is usually regarded as having more reliable and higher quality print performance. To examine the PME printer's performance, common filamented materials spanning a range of print temperatures and materials properties were selected for testing: PLA, PCL, and TPU (Table S2, Figs. S3(A–E), Fig. S4(A–E)). Print parameters for the PME and FFF were then determined empirically (Table S3). We compared the dimensional accuracy, print quality, and mechanical properties of the printed test specimens across the two related platforms.

The print accuracy along each print axis (x,y,z) was determined by printing 20 × 20 × 20 mm^3 cubes from PCL and measuring the length across each axis using digital calipers (Fig. S6). A 0.6-mm nozzle was used for the prints. The PME average error percent across the x-, y-, and z-axis were 0.5 ± 0.2 %, 0.5 ± 0.3 %, and 1.0 ± 0.7 %, respectively, which corresponds to no larger than 0.5 mm of error in each print (Table S4). The error from the PME was found to be comparable to that of same cube design printed with a commercial FFF printer using the same nozzle size (average error percents across the x-, y-, and z-axis of 0.7 ± 0.1 %, 0.5 ± 0.2 %, and 0.7 ± 0.2 %, respectively; Table S4).

Print quality between FFF and PME printed parts were evaluated using scanning electron microscopy (SEM) and optical images of test prints. SEM images along the z-axis of printed parts for PCL reveal exceptional layer line consistency, smooth finish of the extruded layers, and little difference compared with parts printed using a FFF printer (Fig. 4A). Comparison of SEM images for layer line quality of FFF and PME printed parts for TPU and PLA also show near indistinguishable layer line quality and consistency between PME and FFF printed objects (Fig. S7). Additionally, optical images display the print quality achievable with the PME printer (Fig. 4A–D). Standard benchmark models of a tugboat and an octopus were printed using PCL and TPU, respectively (Fig. 4C, D). Overall, the printer was able to handle challenging geometric features including overhangs, edges, curves, and pillars. Additionally, various infill geometries (e.g., grid, cubic, lines, gyroid, triangles, cubic subdivision, trihexagon, octet, quarter cubic) and infill percentages (triangles at 15 %, 30 %, 45 %, and 60 %) were printed (Fig. 4B, Fig. S8, Fig. S9). The PME printer performed each of these operations without issue.

We assessed mechanical properties through quasistatic tensile testing of ASTM D638 type V samples printed using PME and FFF printers using PCL, TPU, and PLA as build materials (Fig. S10). The Young's modulus, elongation to break, and yield stresses were compared (Fig. 5, Table 1). Values obtained from each print method were found to be comparable with one another. (Fig. 5, Table 1). We found that PCL printed with the PME had a higher elongation at break than parts printed with the FFF printer (Fig. 5, Table 1). We hypothesized that the differences in the PCL samples were either due to print speed (the PME prints slower than FFF printers, Table S3) or to the grinding of polymer into powder. Notably, polymer processing conditions and polymer mastication have each been shown to alter the properties of the resulting material[42,43]. To better understand this, we printed PCL on a FFF printer at the same print speeds as the PME printer, which resulted in more similar properties between the printed test samples (Fig. 5A, Table 1). Additionally, the tested samples were reprocessed into powder and reprinted into tensile test specimens (Fig. 5A). Reprocessing resulted in reduced the Young's modulus and tensile strength of the parts, but retained comparable elongation at break (Fig. 5, Table 1). Collectively, the tests establish that PME printed parts can achieve the same mechanical properties as those produced by FFF printers, but the print parameters and effects of polymer mastication should be considered.

5. Case studies

An advantage of PME printers is the potential to expedite materials research due to the simplicity in materials preparation and mixing. We hope the accessibility and robustness of this system will promote breakthroughs in sustainability and material discovery through rapid testing of a variety of material systems. To illustrate this, several model systems were explored, demonstrating the printing of recycled commodity plastics, polymer composites, polymer blends, and a functional mechanophore-containing polymer.

5.1. Commodity plastics

Mechanical recycling is a widely employed method for recycling plastic solid waste, but comes with challenges including contamination of feedstock, reduced mechanical properties, and separation of multilayered plastics[44,45]. PME printing offers a mode for the study and

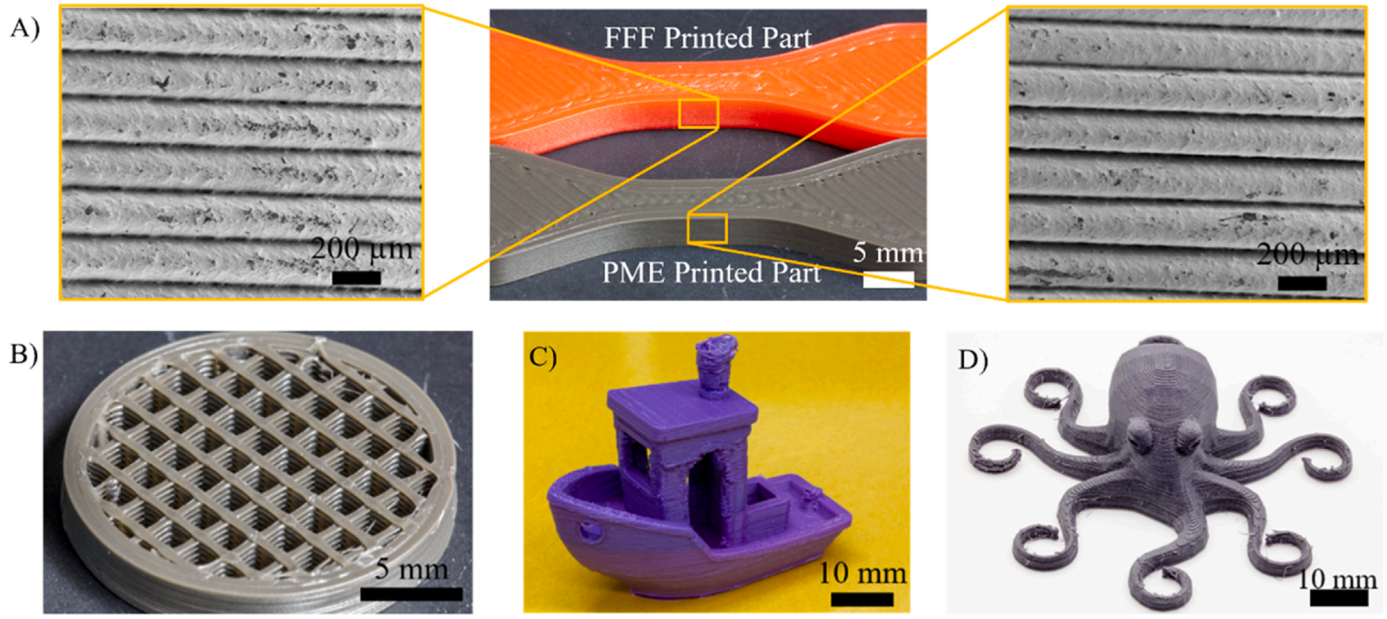


Fig. 4. (A) SEM images showing the layer lines of PCL printed with the PME and FFF printer (B) Optical image showing a single infill of a PME printed object (C) PME printed benchmark model (D) PME printed octopus.

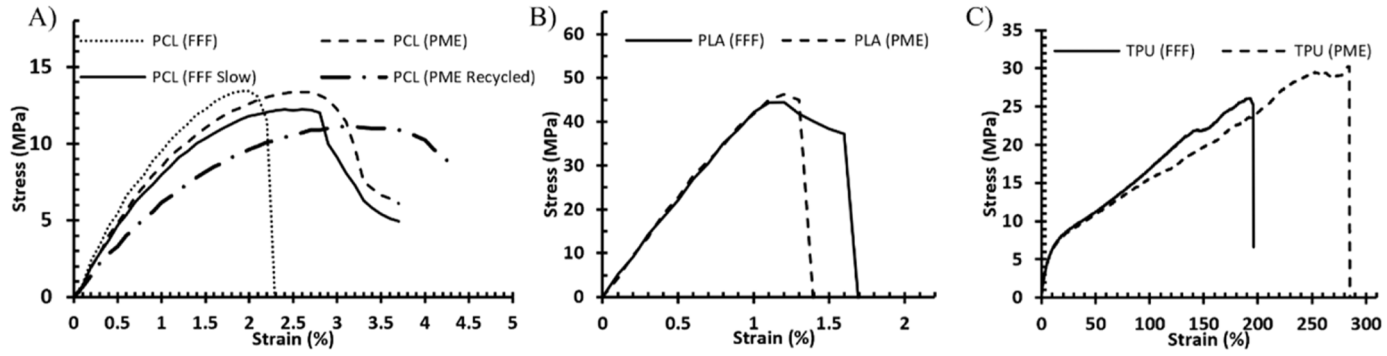


Fig. 5. Uniaxial tensile testing data showing the stress-strain behavior of representative samples from three replicate trials (Fig. S10) for ASTM D638 type V tensile samples printed from A) PCL printed with FFF (dotted), PME (dashed), FFF at slower print speeds (solid), and PME recycled from previous prints (dash dot). B) PLA printed with FFF (solid) and PME (dashed). C) TPU printed with FFF (solid) and PME (dashed).

Table 1
Summary of the tensile properties of printed PCL, PLA, and TPU ASTM D638 type V samples from PME and FFF printers.

Samples[a]	Young's Modulus (Mpa)	Yield Stress (Mpa)	Elongation at Break (%)
PCL (FFF)	1240 ± 90	14.7 ± 0.2	2.4 ± 0.2
PCL (PME)	1120 ± 60	14.3 ± 0.4	3.7 ± 0.3
PCL (FFF Slow)	1030 ± 40	13.5 ± 0.2	3.5 ± 0.3
PCL (PME Recycled)	900 ± 100	12.4 ± 0.6	4.0 ± 0.5
PLA (FFF)	4600 ± 300	43 ± 7	1.9 ± 0.2
PLA (PME)	4700 ± 200	49.6 ± 0.8	1.43 ± 0.06
TPU (FFF)	61 ± 5	8 ± 1	200 ± 30
TPU (PME)	57 ± 3	9.2 ± 0.1	260 ± 40

[a] Averages and standard deviations represent the results from three replicate experiments.

implementation of the mechanical recycling process on a reduced scale and turns recycled polymer directly into printed objects. To demonstrate mechanical recycling of commodity plastics, high density polyethylene (HDPE) lids taken from laboratory scintillation vials and from plastic soda bottles were processed into powder. Notably, no additional pretreatment steps were used, and the HDPE comes from two different sources. The resulting powder mix, as expected, exhibited two melt and crystallization temperatures (Fig. S3E). The powder HDPE can easily be recycled into additional printed objects using PME printing, effectively demonstrating material recycling as a potential use (Fig. 6A, Table S3). Even this simple demonstration is encouraging and creates prospects for studying the recycling of commodity plastics from multiple material streams or the reprocessing of multilayered plastics.

5.2. Polymer composites

The relative narrow range of FFF 3D printable materials and resulting properties has motivated the development of composite materials [46–49]. These polymer composite materials give access to properties typically not available from accessible polymer materials as well as properties that can even outperform those of the individual components [50]. Often, the processing of composite materials is a challenge, which motivates direct formation of composite objects during the AM printing process. To show the ease of creating novel composite systems, strontium aluminate, a common phosphor found in glow-in-the-dark products, was mixed with TPU in a ratio of 80:20 by weight (TPU:strontium aluminate). The composite system could then be printed into objects

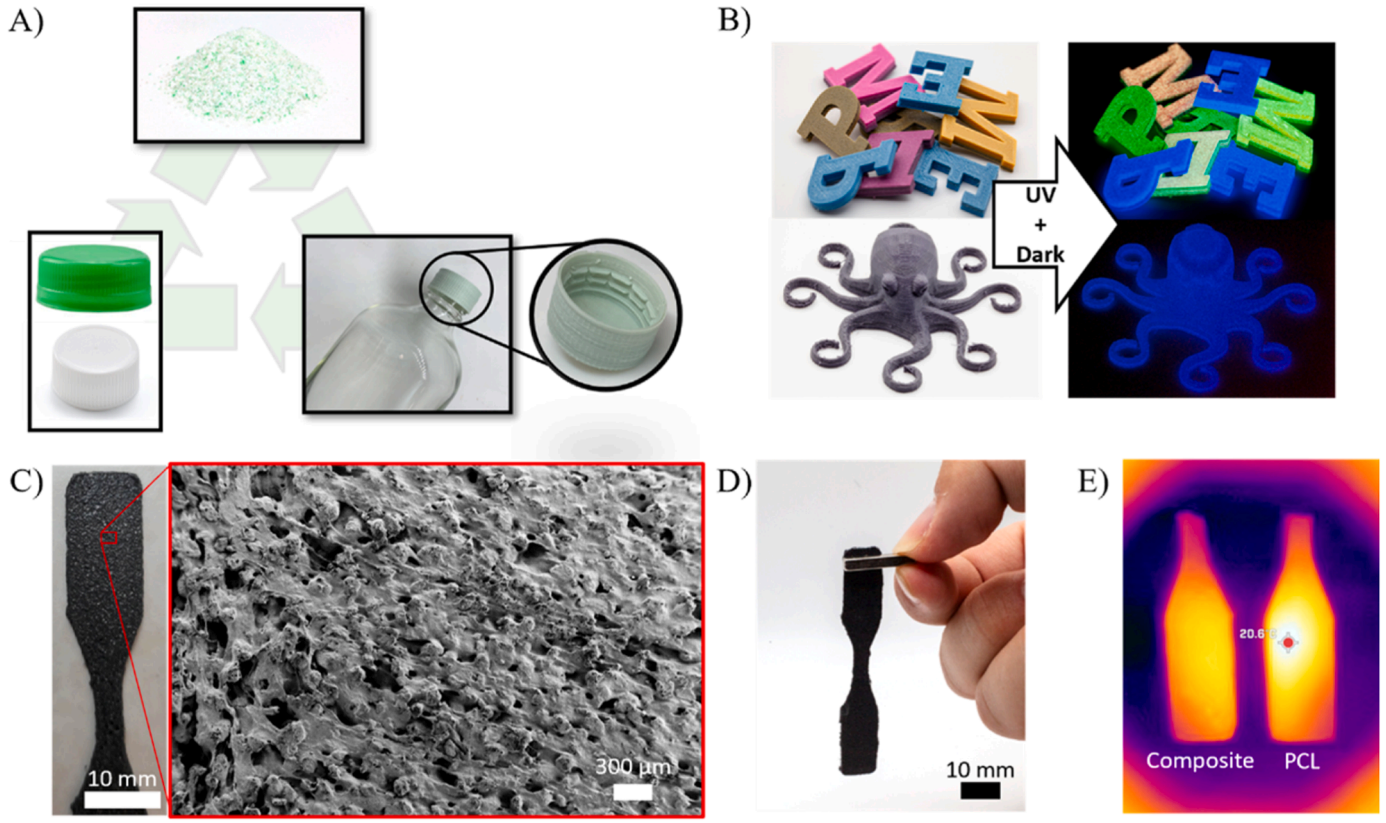


Fig. 6. A) Transformation of combined waste plastic back into useable items. B) Printed strontium aluminate composites and demonstration of their glow in the dark properties. C) Printed iron oxide and PCL composite along. Corresponding SEM shows the iron oxide particles along the surface of the printed part. D) Iron oxide and PCL composite suspended using a magnet. E) IR thermal camera image of iron oxide PCL composite being cooled next to a pure PCL part.

with various shapes and geometric complexities, and the parts were phosphorescent after exposure to UV light (Fig. 6B).

Separately, iron oxide could be mixed with PCL and printed to form composites (Fig. 6C–E). Printed composites contained a matte-like and rough surface that, upon closer inspection by SEM, shows the incorporations of the iron oxide throughout the thermoplastic matrix (Fig. 6C). Increased loadings of inorganic fillers often require increased nozzle diameters. We briefly explored printing with nozzle diameters ranging from 0.4 to 1 mm using PCL (Fig. S11). Print parameters only required minor alteration to accommodate for the changes in pressure drop and volumetric flowrate for the different nozzles (Table S5). We also determined that we were able to print compositions consistent with the mixture loadings of iron oxide and PCL. By loading a 60 wt% iron oxide 40 wt% PCL mixture into the hopper we were able to print objects and used TGA to determine the composition of the part (Fig. S12). TGA showed good agreement with loaded mixture as 62.2 % of material was left after thermally removing the PCL. We were also able to reach up to 80 wt% loadings of iron oxide in PCL using a 0.8 mm nozzle with compositions confirmed by TGA (Fig. S12). Although there are many uses for thermoplastic composites, the ability to reach 80 wt% iron oxide loadings reveals potential for printing metals and ceramics where debinding and sintering would be needed to form the final part. Notably, the printed parts exhibited ferromagnetic properties and could be attracted by standard bar magnets (Fig. 6D). We also observed an increase in the overall thermal conductivity of the parts. Additionally, comparison of PCL objects with and without iron oxide (~50 wt% of iron oxide) resulted in differences in their cooling rate with the composite part cooling faster than the pure PCL part (Fig. 6E).

5.3. Polymer blends

Formation of polymer blends can be a powerful method to obtain materials with improved properties and interesting phase behavior [51, 52]. PME printing offers a mode for simple implementation and study of these properties. For example, by printing a PCL/TPU polymer blend, we achieved thermally responsive shape memory materials [53,54]. These thermally responsive shape memory materials can be printed into a permanent shape which, when heated above a thermal transition temperature (T_{trans}), can be deformed and subsequently cooled below T_{trans} locking in the temporary shape until being reheated above T_{trans} to return the object to its permanent form (Fig. 7A). In the case of PCL/TPU blends, the shape memory properties are derived from the distinct thermomechanical properties of the individual polymers. Dynamic mechanical analysis (DMA) of each individual polymer reveals neither material alone exhibits shape memory behavior within the temperature range measured (Fig. S13). TPU maintains a steady storage modulus and tan δ while PCL has a drastic decrease in storage modulus and increase in tan δ as it forms a polymer melt where the viscous properties begin to dominate and shape fidelity is lost (Fig. S13). Contrasting these behaviors, a 40/60 by weight blend of TPU/PCL allows for a more moderate decrease in storage modulus that plateaus resulting in only a slight increase in the tan δ (Fig. S13). This softening allows for the blended polymers to be deformed above the melting temperature of PCL, locked into place by cooling and reforming of the stiff PCL crystalline domains, and then returned to the original shape through additional heating (Fig. 7A). This process is shown by printing a star, heating and deforming, allowing it to cool to retain its temporary shape, and then reheating to form the original star shape (Fig. 7B). Additionally, this shape transformation can be fairly rapid when used in thin designs where heat transfer can be fast. This was shown through printing of a

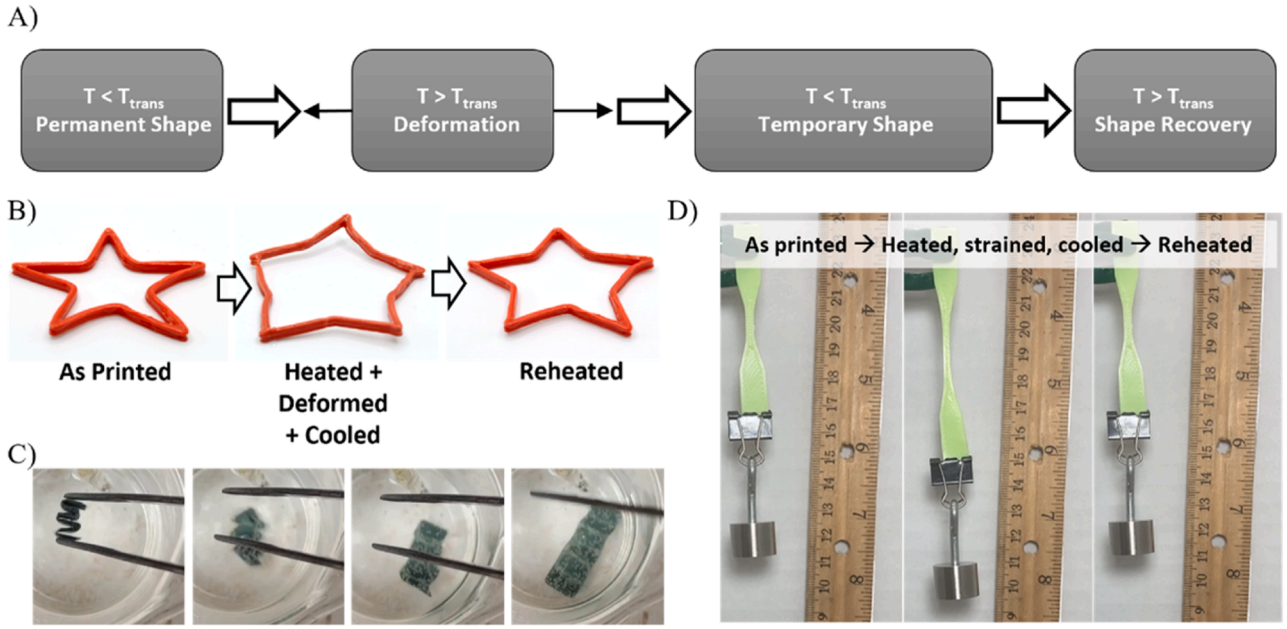


Fig. 7. A) Demonstration of thermally responsive shape memory material below and above it transition temperature (T_{trans}). B) 3D printed star from a PCL/TPU blend that allows for 3D printing (left), heating and deforming of material into a temporary shape (middle), and then reheating to reform original shape (right). C) Demonstration of rapid shape transformation of PCL/TPU blend printed in a panel design with unfolding shown in images from left to right as material is placed into 60 °C water. D) Ability to store elastic energy using thermally responsive shape memory materials.

panel design that can be folded and fixed into shape, but when reheated transforms back into its original shape in seconds (Fig. 7C). Furthermore, the shape memory properties of the polymer blend can be tuned by altering the composition of the TPU/PCL which was explored by bending tests (Fig. S14, Table S6). The 20/80 and 80/20 mixtures of TPU/PCL show lower shape memory properties than mixtures of 40/60 and 60/40 TPU/PCL. This is likely due to minority components being unable to form continuous networks. Interestingly, these polymer blends can also be used to store elastic energy to be released later upon thermal heating. By heating a printed part from a 60/40 mixture of TPU/PCL to 60 °C, straining the sample, and then cooling the sample, elastic energy is stored. The release of this stored energy is shown by attaching a 20 g weight to the sample and reheating (Fig. 7D).

5.4. Specialty polymers

In many instances, researchers are interested in 3D printing novel or functional polymers that are not commercially available and may be difficult to synthesize at multigram scales. In these instances, the PME may be an appropriate platform for printing and exploring the properties of these materials. One exciting avenue is the printing of mechanophore containing polymers. This was explored by the synthesis of PCL containing a chain centered spiropyran, P1 (Fig. 8A). Under either mechanical force or UV light the spiropyran undergoes isomerization to its merocyanine form (Fig. 8A). Parts can be printed from P1 into arbitrary geometries (Fig. 8B). When printed objects are exposed to UV light, they undergo a stark change in color (Fig. 8C). Mechanical force can also be

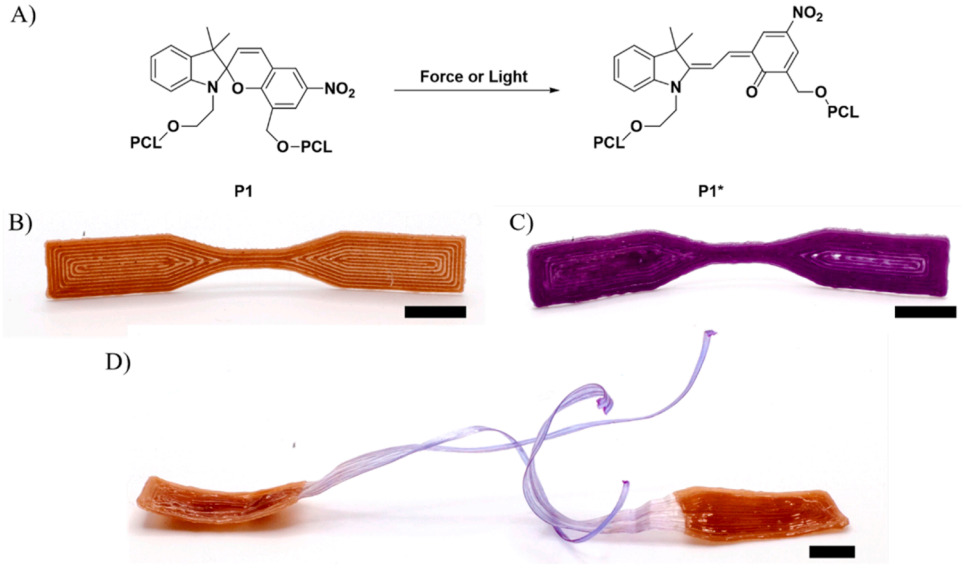


Fig. 8. A) Generalized depiction of mechanical and UV induced isomerization of Spiropyran containing polymer, P1, to its merocyanine form, P1 * B) Part printed from P1C) Image of the P1 printed part after exposure to UV light D) Image of P1 after being manually stretched. All scale bars indicate 10 mm.

used for strain induced mechanophore activation (Fig. 8D).

6. Discussion

The development and application of the open-source PME 3D printer provides an accessible, low-cost, and versatile PME printing system. The PME printer we introduce is particularly beneficial for providing access to chemical and material spaces that are incompatible with filament-based and compounding materials, making it impactful for research and development purposes.

Our findings demonstrate that the open-source PME printer can achieve print quality comparable to commercial FFF systems, particularly in terms of resolution, appearance, and mechanical properties. This is a noteworthy achievement considering the intrinsic complexities of PME systems compared to FFF. The PME system's ability to handle a broad spectrum of materials, including recycled plastics, polymer blends, composites, and specialty polymers, underlines its versatility and potential for broader application in material science research.

The PME printer's capacity to work with a diverse suite of materials, ranging from thermoplastic polymers to complex composites, opens new avenues for material exploration. The successful printing of materials with special properties like phosphorescence, ferromagnetism, shape memory, and mechanochromism is particularly significant. These capabilities indicate the potential for developing advanced materials with customized functionalities tailored for specific applications, ranging from biomedical devices to smart materials.

Conclusion and future work

While the PME printer marks a significant step forward, challenges remain, particularly in achieving consistent material delivery over longer timescales and handling a wide range of material granule sizes. Future work could focus on refining the printer design for even more consistent powder delivery and exploring novel material systems, such as compatibilized multi-stream plastics and organic-inorganic composites. Additionally, further development of the printer to facilitate in-line mixing of materials could pave the way for smoother material gradients and multi-material printing.

In conclusion, the open-source PME 3D printer offers a promising platform for the rapid prototyping and testing of novel material systems. Its accessibility, cost-effectiveness, and versatility make it a valuable tool for both academic research and industrial applications. By bridging the gap between material availability and printing technology, the PME system has the potential to significantly accelerate the exploration and development of new materials and their applications in various fields.

CRediT authorship contribution statement

Chin Kyle C. H.: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Boydston Andrew:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Miao Xinyu:** Writing – review & editing, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.addma.2024.104014.

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